DSA Day 44: Storage Management + Review

Book: Data Structures Through C in Depth - Ch10 §10.1-10.7 · pp 492-508 | Code in Python

Overview

Allocators pick a free block (first/best/worst fit); fragmentation wastes space; buddy systems and garbage collection (reference counting, mark-and-sweep) manage memory. Then review the whole book.

Key concepts to master

- first/best/worst fit
- fragmentation
- boundary tag, buddy system
- reference counting, mark & sweep

Python implementation

```
def first_fit(blocks,size):
    for i,b in enumerate(blocks):
        if b>=size:
            blocks[i]-=size; return i
    return -1
```

Complexity

Allocation strategies vary; GC amortised.

Common pitfalls

- External vs internal fragmentation confusion
- Reference counting misses cycles

Practice

- Code a first-fit allocator
- Re-code your weakest data structure